

5 Case study

MS1FA

Files upload

Select parameters

Output

Documentation

Feature table file

1

Choose a feature table file (.csv)

Browse... Demo file selected

2

Select columns

AR351_A, AR351_B, AR351_C, Cipro_A, Cipro_B, Cipro_C, Coumermycin_A, Coumermycin_B, Coumermycin_C, CN.DM.861_A, CN.DM.861_B, CN.DM.861_C, Levo_A, Levo_B, Levo_C, Lome_A, Lome_B, Lome_C, Novo_A, Novo_B, Novo_C, untreated_A, untreated_B, untreated_C

☐ Show feature table ☒ Show demo feature table

3

File	File Description	Download
☒ PA14 XCMS feature table	This is a LC-ESI-MS example dataset from an untargeted metabolomics experiment investigating <i>Pseudomonas aeruginosa</i> perturbed with different antibiotics. The feature table was generated using XCMS.	Download
☐ STM16 XCMS feature table	This LC-ESI-MS dataset was generated from a mixture of 16 pure standards spiked into the PA14 samples. The feature table was generated using XCMS.	Download
☐ STM16 MZmine feature table	This LC-ESI-MS dataset was generated from a mixture of 16 pure standards spiked into the PA14 samples. The feature table was generated using MZmine4.	Download

☐ View demo feature table [Select demo](#)

4

MS2 file

4

Choose a MS2 file (.mzXML, .mzXL or .mgf)

Browse... Demo file selected

5

MS2 File	MS2 File Description	Download
☒ PA14 MS2 file	DDA MS2 data in mzXML format of a pool sample corresponding to the PA14 feature table.	Download
☐ S116 MS2 mzXML file	DDA MS2 data in mzXML format of a pool sample of 16 standards corresponding to the STM16 feature table.	Download
☐ STM16 MS2 mgf file	MGF file generated from DDA MS2 data of the STM16 mixture using MZmine4.	Download

☐ View demo MS2 table [Select demo](#)

6

Library file

6

Choose a metabolite target list

Browse... Demo file selected

7

Metabolite Target List	File Description	Download
☒ PA14 Target List	A target list compiled from the <i>Pseudomonas aeruginosa</i> Metabolome Database (PAMDB).	Download
☐ STM16 target file	A target list of 16 standards.	Download

☐ Show metabolite target list [Select demo](#)

8

Neutral loss file

8

Choose Neutral Loss File

Browse... Default file selected

9

ESI-MS Adducts

9

Choose adducts file

Browse... Default file selected

☐ Show Data Table

Figure S8. Screenshot of MS1FA files upload.

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To demonstrate the functionality of MS1FA we use the provided PA14 example data. The process begins with uploading the feature table file (Section 1). Once the file is uploaded, a dropdown menu will appear (Section 2) to select the columns representing sample intensities. For replicated samples, please use consistent naming suffixes like '_A', '_B', '_C' or '_1', '_2', '_3'. In Section 3, three demo feature table files—PA14 processed via XCMS, StM16 processed via XCMS and MZmine—are available for users to test and download. Next, in Section 4, users upload an MS2 file. MS1FA supports .mzXML or .mzML files from a pool sample measured with DDA MS2, or a .mgf summary file from MZmine generated from multiple MS2 files. We provide two demo MS2 files for testing in Section 5: one is a .mzXML file for the PA14 dataset, one is a .mzXML file for the StM16 dataset, and the other is a .mgf summary file for the StM16 dataset generated from multiple MS2 files. In Section 6, users need to upload a metabolite target list in csv-format. Alternatively, it is also possible to upload spectral libraries in .msp format or Bruker .library format, which are parsed to extract the information to generate a target list. The target list must include essential information such as the metabolite name and chemical formula. In Section 7, we provide two example target lists: one for the PA14 dataset, which includes metabolites compiled from the *Pseudomonas aeruginosa* Metabolome Database (Huang et al. 2018) and a target list for the StM16 dataset, containing 16 metabolites. In section 8 and 9 default files which contain tables for neutral losses and adducts, respectively, are uploaded automatically. Upload of user-generated tables is also possible.

The screenshot displays the MS1FA web interface with the following sections:

- Section 1: Filter the feature table**
 - Minimum RT value (in seconds): 60
 - Maximum RT value (in seconds): 1200
 - RT Range in seconds: A slider from 0 to 1,600 with markers at 0, 160, 320, 480, 640, 800, 960, 1,120, 1,280, 1,440, and 1,600. The range is currently set from 60 to 1200.
 - ☐ Filter ISF features for compound identification
- Section 2: Check isotopes and multiple charge states**
 - ☒ Check C13 isotopes and multiple charge states
- Section 3: Correlation method**
 - Correlation method: pearson
- Section 4: Correlation threshold**
 - Correlation threshold: 0.8
- Section 5: Retention time threshold (in second)**
 - RT threshold for correlation: 3
 - RT threshold for metabolites identification: 30
 - RT threshold for precursor feature matching: 20
 - RT threshold for MS2 feature matching: 2
 - RT threshold for NL feature matching: 2
 - RT threshold for adducts feature matching: 2
- Section 6: Ion polarity and primary ions**
 - Ion polarity: pos
 - Primary ion: [M+H]⁺ [M+Na]⁺
- Section 7: ppm**
 - ppm for exact m/z matching: 5
 - ppm for precursor m/z matching: 5
 - ppm for MS2 m/z matching: 10
 - ppm for neutral losses m/z matching: 5
 - ppm for adducts m/z matching: 5
- Section 8: m/z difference tolerance**
 - m/z difference tolerance for exact m/z matching: 0.002
 - m/z difference tolerance for precursor m/z matching: 0.002
 - m/z difference tolerance for MS2 m/z matching: 0.005
 - m/z difference tolerance for neutral losses m/z matching: 0.002
 - m/z difference tolerance for adducts m/z matching: 0.005

A "Run" button is located at the bottom right of the interface.

Figure S9. Screenshot of MS1FA select parameters.

Before running MS1FA, several parameters need to be set. First, the feature table is filtered by retention time (RT) to exclude unwanted features (Section 1). The default filter range, set from 60 to 1200 seconds, is designed to suit the demo datasets by removing the calibration peak and the gradient's wash phase. And a checkbox is provided if the user wants to skip the metabolite annotation when the feature is already identified as a MS2 matched in-source fragment. For ^{13}C isotope annotation, the checkbox has to be selected as default (Section 2). The next section involves choosing the measure of the correlation from the dropdown menu (Section 3) and setting the correlation coefficient threshold (Section 4), which defaults to 0.8. After that, the RT thresholds for the acceptable time windows have to be configured (Section 5): 30 seconds for metabolite identification via exact matching (if RT is provided in the target list), 20 seconds for precursor ion matching from MS2 data to the MS1 feature table, 2 seconds for matching MS2 features to precursor features in the MS1 table, and 2 seconds each for matching pairwise features to the neutral loss and adduct tables. Additionally, select the ion polarity and primary ions (Section 6), the default is positive polarity with $[\text{M}+\text{H}]^+$ and $[\text{M}+\text{Na}]^+$ ions, suited for the demo datasets. Next, we specify the ppm thresholds (Section 7) for exact mass matching (default 5 ppm), precursor ion m/z matching (default 5 ppm), MS2 fragment m/z matching (default 10 ppm), neutral

loss m/z matching (default 5 ppm), and adduct m/z matching (default 5 ppm). Finally, we set the m/z difference tolerance thresholds (Section 8): 0.002 for exact mass matching, 0.002 for precursor ion and 0.005 for MS2 m/z matching, and 0.002 for neutral loss and 0.005 for adduct m/z matching. After setting all the parameters, click “Run” to start calculation. The default settings are well suited to process the demo data sets.

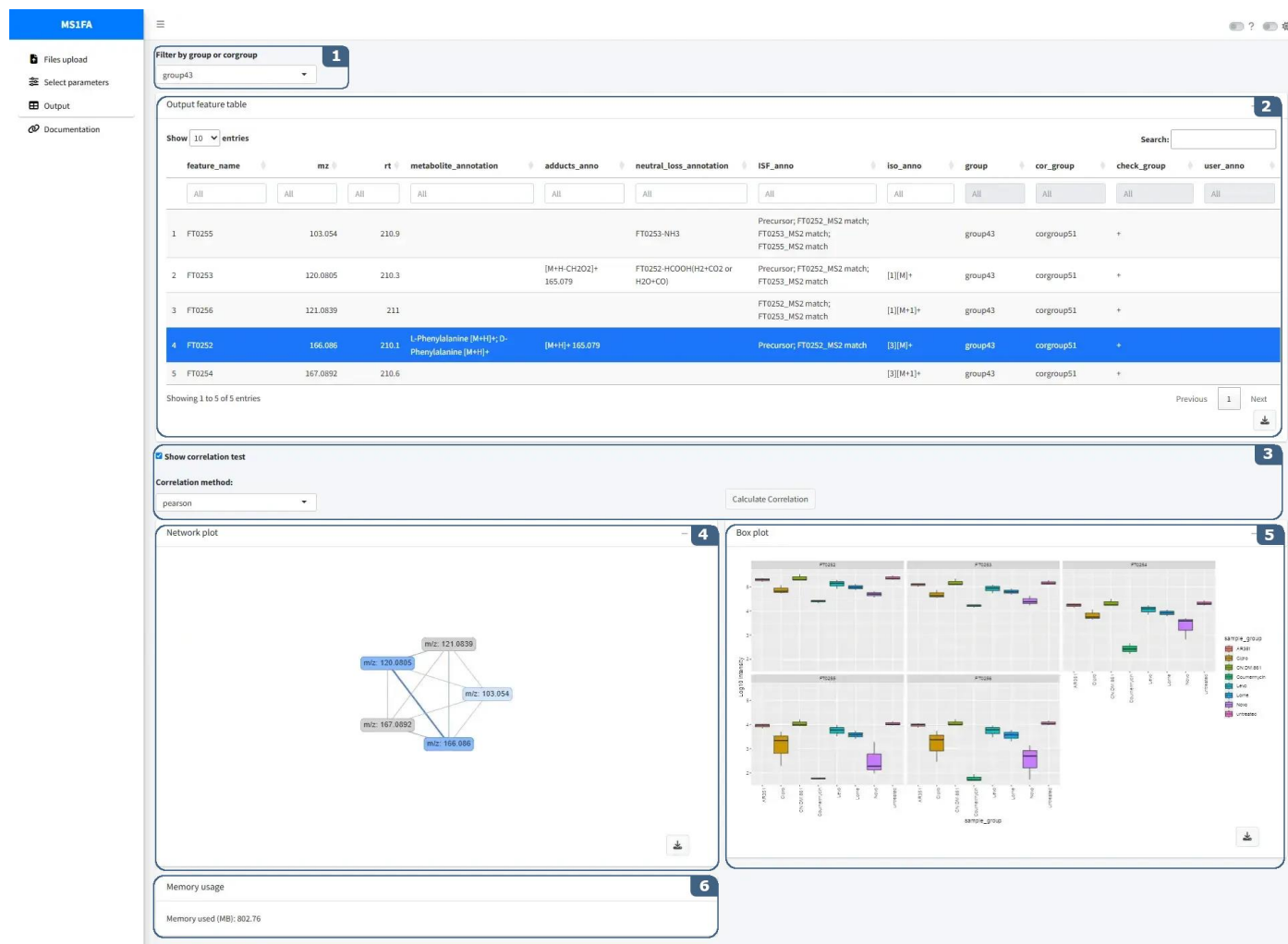


Figure S10. Screenshot of MS1FA output.

The MS1FA output includes an interactive feature table with annotations (Section 2). The first column **feature_name** contains the unique feature identifier, the second **mz** and third column **rt** display the mass-to-charge ratios and the retention time in seconds respectively. For the highlighted feature, the unique identifier is FT0252 with an m/z of 166.086 and a retention time of 210.1 seconds. The column **metabolite_annotation** shows the annotation via exact mass matching (and RT matching if available) using a target list. Matching of FT0252 to the PAMDB target list yields two matches L-Phenylalanine and D-Phenylalanine. Adducts annotation and neutral loss annotation takes place by matching to the adduct and neutral loss annotation tables, shown in the columns **adducts_anno** and **neutral_loss_annotation**, respectively. For our example, the adduct is correctly identified as [M+H]⁺ and the neutral monoisotopic mass is also calculated as 165.079. The next column shows the annotation of putative neutral losses

generated by comparing each pairwise mass difference within a defined RT window to the neutral loss table. For FT0253 ($m/z = 120.0805$), a neutral loss of HCOOH was found in relation to FT0252 ($m/z = 166.086$). This neutral loss is verified in the next column **ISF_anno**, which shows the in-source-fragment (ISF) annotation via matching to MS2 data. For FT0252, we find a FT0253_MS2 match, which means that the 120.0805 m/z (FT0252) is found as a fragment ion in the MS2 spectrum with the precursor 166.086 m/z (FT0253). The column **Iso_anno** shows the annotated isotopes. The number in the first square brackets shows the isotope-group-id and $[M + 1]$ is the first isotopic peak for the monoisotopic peak $[M]$. The column **group** shows the grouping of related features, as explained in the Supplementary Information. The results of the grouping by intensity pattern method are shown in the column **cor_group**. The column **check_group** shows “+” when the most overlapping assignment is identified by both groups. The last column **user_anno** can be used by the user to fill in comments and/or own annotations that will be exported with the annotated feature table as well.

Section 3 introduces the checkbox “Show Correlation Test”, which allows users to view correlation statistics. When exactly two rows are selected in the feature table, a dropdown menu appears, enabling the user to choose from different correlation methods. After clicking “Calculate Correlation”, a modal dialog displays the results from R’s `cor.test()` function.

When a row in the feature table is selected, a corresponding interactive correlation network plot is generated (Section 4), displaying feature names, annotations when hovering over nodes, and correlation coefficients along with m/z differences when hovering over edges. In the network plot, a gray node color indicates that the feature is recognized as an isotope. A dark blue node signifies that a metabolite has been annotated, while a light blue node indicates that the feature is not identified as a metabolite. Edges represent correlation values, m/z differences, and neutral loss annotations with thicker edges highlighting detected neutral losses.

Similarly, selecting a row in the feature table allows users to generate a box plot (Section 5), offering insights into the correlation related of features across different samples. The titles of each box plot is the name of feature. The classes are the samples groups and the values are the $\log_{10}$ transformed feature intensity value.

In Section 1, a dropdown menu allows filtering of the output feature table by group index or corgroup index. Section 6 shows the memory usage.